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Timestamping and synchronization method

This article provides an overview of the iMotions timestamping method – how exactly does the iMotions software synchronize the data-streams of multiple devices? The article further links to tools which enable the user to take control over some aspects of the timestamping method.

Imagine the following two scenarios.

Scenario 1 – You are part of a global-scale company, which is about to launch a world-wide media campaign for a new product; it includes printed advertisements, video commercials, and web-content. It is your job, as a neuromarketing researcher, to predict whether the campaign is going to be successful. To this end, you plan a study using eye-tracking ("Does the material attract attention?"), GSR ("Is the material emotionally engaging?"), Facial Expression Analysis ("Are there signs of genuine interest?"), and EEG ("Does the material evoke approach-tendencies?").

Scenario 2 – You are an academic researcher, who received funding for a health-project in patients with Autism. Specifically, you plan a study wherein Autism patients are exposed to emotionally expressive faces (e.g., the face of a person showing 'joy', 'disgust' or 'anger'). You intend to use a combination of eye-tracking ("Do Autism patients attend to the same parts of a face as normal controls do?") Facial Expression Analysis ("Do Autism patients mimic emotional faces in the same way as normal controls do?"), and EEG ("Do Autism patients process emotional faces in the same way as normal controls do?").

In both scenarios, you will have been told that the iMotions software-platform is the perfect solution for your research – for the specific reason that it allows you to collect data from many different types of biometric sensors within one and the same software. As such, iMotions helps you to easily combine the insights from very different technologies, in order to ultimately obtain a powerful and rich answer to your specific research question.

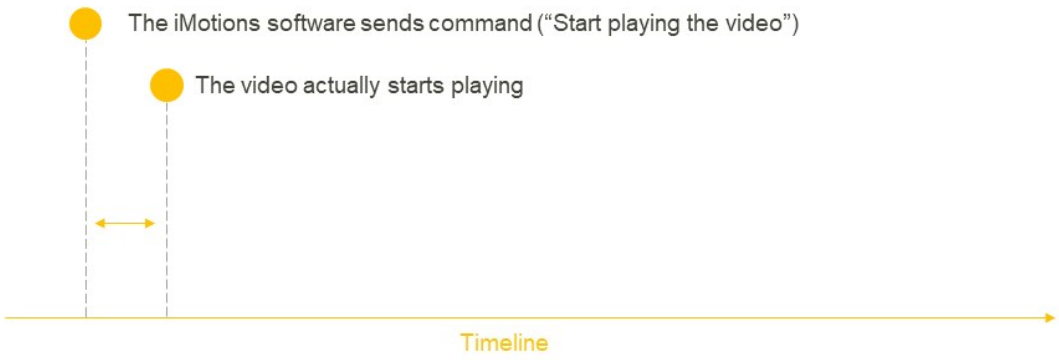
However, one element is essential to this ability to 'combine the insights from very different technologies': *synchronization*. More precisely, the synchronization of all data-streams on a common timeline.

What does that mean: "synchronization"?

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‘Synchronization’ is a tricky thing. Or rather, ‘*perfect synchronization*’ is a tricky thing.

To illustrate this point, first, consider what it takes to present ‘stimuli’ on a screen: an image, a video, a website.



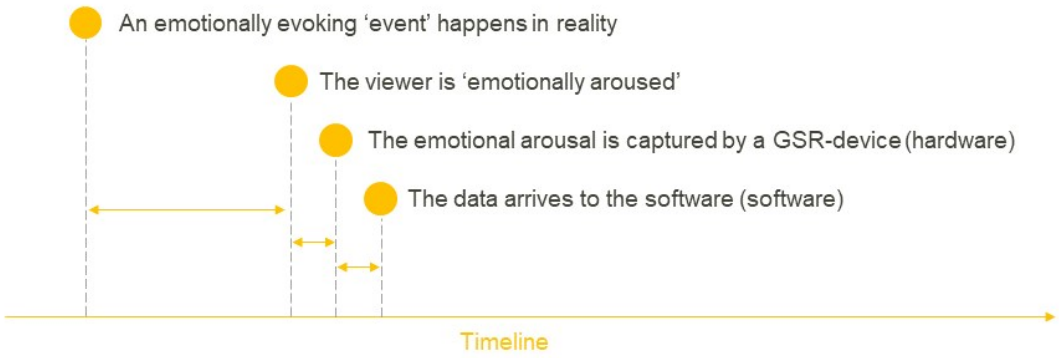
It always starts by sending a command – a coded message sent from the iMotions software to, for instance, a video player: “*please start playing the video*”. But the time of sending the command is not the time that the video actually starts playing. It always takes a few tens, or even hundreds, of milliseconds until the video player is able to execute the command. It is therefore not enough to know when the command is being sent from iMotions. We also need to know the latency until the command is being executed by the video player (see below: the difference between ‘StartSlide’ and ‘StartMedia’).

A similar latency exists in the opposite direction, that is, in the recording of biometric data. Now that the video is playing on screen, imagine that this video contains an ‘emotionally arousing’ event. For instance, imagine that the video contains a funny joke, which evokes a response captured by GSR: an ‘arousal peak’. Even though the response is directly *evoked* by the event in the video, it does not necessarily *co-occur* with the event in the video. In fact, it may take a few seconds until this response is, indeed, notable in the data.

Why is that?

Several factors contribute to this latency.

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The first factor is the nature of the biological response itself (and this is immediately also the factor that, in most cases, accounts for the largest portion of the latency). Emotional responses are not instantaneous. In order to respond to an emotional event the human brain first needs to process the information. And this takes a bit of time. Depending on the specific type of response – whether it is a saccade (i.e., a goal-directed eye-movement), a smile, an arousal peak (in GSR), or an increase in alpha activity (in EEG), it may take between a few hundred milliseconds, and up to a few seconds, before the biological response happens.

A second factor relates to the hardware used. Also the registration of the 'emotional responses' by the hardware is not instantaneous. It takes a bit of time for the hardware to capture the response – a latency in the range of a few milliseconds, up to a few tens of milliseconds. If the hardware system itself computes 'metrics' (which are often based on data collected within a somewhat larger time-window), this delay may increase to a few seconds.

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A third factor, finally, relates to the software – more precisely, to the communication between the hardware (e.g., the eye-tracker) and the computer used for data collection (running the iMotions software). Depending on the communication protocol used, it takes a few tens, and up to a few hundreds, of seconds to send over data from the hardware to the iMotions software. Some devices communicate through a wired connection, which is the most reliable type of communication (without data loss) but it is not necessarily fast; other devices communicate using a wireless connection, for instance, over Wi-Fi or Bluetooth. Of these, Wi-Fi is quite reliable and the communication is fast, but these latencies also vary a lot (between 5 milliseconds and several hundreds of milliseconds). Bluetooth, on the other hand, is more susceptible to data loss, and the communication is quite slow, but with less variation (between 25 and 50 milliseconds). Moreover, many devices do not deliver the data sample-by-sample, but in the form of data packages – whereby a larger number of samples are delivered at the same time. In that case, the last sample of the package might correspond to data obtained a few milliseconds earlier, whereas the first sample of the same package corresponds to data obtained a few hundreds of milliseconds earlier.

For the live visualization of data yet another factors comes into play – a factor which, essentially, is the same as the latency discussed already for presenting ‘stimuli’ on screen. Namely, also after data samples have arrived to the iMotions software, it still takes a bit of time before the data can be displayed in the live visualizer. This factor, notably, only affects the visualization of the data in the live visualizer, but not the data collection itself.

Indeed, quite a number of factors need to be considered in order to obtain a ‘perfect synchronization’. Some of these factors can be addressed at the level of the software, while other factors cannot be addressed by the software.

So, what exactly does the iMotions ‘synchronization’ entail?

The iMotions method

The iMotions method includes a combination of strategies which, together, minimize the various types of latencies that exist. It is important to realize that, just as much as the iMotions software cannot speed up the response of the human brain, there is no magical way in which the iMotions software can be *more precise* than what is delivered by the hardware – but we do offer solutions which compensate for the latencies introduced by the hardware.

Timestamping

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For the ‘timestamping’, the iMotions software relies on the internal clock of the hardware. However, different hardware systems report these timestamps in different ways. Some devices report actual ‘timestamps’ – typically in seconds or milliseconds – while other devices report ‘sample numbers’ (sample 1, sample 2, etc.). The iMotions software therefore adopts one of two rules for timestamping. Rule 1: if the hardware provides actual ‘timestamps’, the iMotions software distributes the samples according to these timestamps. Rule 2: if the hardware provides ‘sample counts’, the iMotions software calculates the timestamps based on the device’s sampling rate. For instance, if the device samples data at 200 Hz (‘200 samples per second’), the samples will be distributed equally, using intervals of 5 milliseconds. This means, in both cases, that the timestamps in the iMotions are equally precise as the timestamps delivered by the hardware itself, without additional delays that may occur due to the respective communication protocols used.

Point-of-synchronization

For the ‘synchronization’ of the devices, the iMotions software uses a point-of-synchronization in the beginning of the recording; all data-samples collected during the recording are relative to this point-in-time. This means that the instantaneous CPU-load at this point in time can play an important role in the size of the offset of the devices. To minimize these offsets, the iMotions software has, again, two strategies in place. The first strategy is to make sure that the data of the different devices is streaming into iMotions already before the start of a recording. A recording can only start if the connection between the hardware and software is successfully established. The second strategy has to do with the computer’s CPU-load. In order to prevent extra delays that may occur due to a high CPU-load at the point-of-synchronization, the point-of-synchronization is chosen in a smart way – at a point in the time *after* all other commands related to the start of the recording have been executed. This ensures that the CPU-load at the point-of-synchronization is reasonably low, and that the system is fully ready to focus on processing data as soon as this point-in-time is exceeded.

Device-specific latency settings

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To account for the remaining offsets of the various devices, the iMotions software works with device-specific latency estimates. These estimates are sometimes based on the documentation provided by the hardware manufacturers, while sometimes they are the results of our own internal testing. This internal testing entailed that, in a controlled fashion, we triggered the sensors using an electrical pulse. In this way, for instance, we estimated that the offset of a Shimmer device is about 28 milliseconds, while, based on the documentation from ABM, we set the offset of to ABM EEG-headset to 40 milliseconds. These device-specific latency-estimates are used by default in the iMotions software. However, if desired, these estimates can be corrected by the user.

Click here to find more information on how to change the default latency settings.

Slideshow events

Also on the stimulus-presentation side, the iMotions software is conscious about the different latencies that exist between sending the command to play a given stimulus on screen and the time that the stimuli actually starts playing. To this end, the iMotions software tracks both the computer's UTC timestamps of when the commands are being sent out of iMotions and the UTC timestamps of when, for instance, the video-player reports that a video is actually playing on screen.



#DATA

Row	Timestamp	SlideEvent	StimType	Duration	CollectionPhase	SourceStimuliNam	EventSource
1	7035.1851	StartSlide	TestImage	30000	StimuliDisplay	Survey_30_1	1
2	7039.6528						
3	7047.4653						
4	7052.4806						
5	7055.2778						
6	7057.1299						
7	7063.0903						
8	7067.1019	StartMedia	TestImage	30000	StimuliDisplay	Survey_30_1	1
9	7070.9028						
10	7078.7153						
11	7086.5278						
12	7086.8645						
13	7089.0534						
14	7094.3403						
15	7102.1528						
16	7109.9653						
17	7117.7778						
18	7121.4222						
19	7125.5903						
20	7133.4028						
21	7136.9144						
22	7141.2153						
23	7149.0278						
24	7155.134						
25	7156.8403						

Latency (~30 ms)

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This is clearly represented in the new ‘Sensor Data’ export-files for iMotions 8.0. These ‘Sensor Data’ export-files include a few new data-columns dedicated to slideshow events, among which the column ‘SlideEvents’. This column contains four types of events: StartSlide > StartMedia > EndMedia > EndSlide. ‘StartSlide’ represent the time that iMotions sends out a command to display a stimulus (‘Slide’) on screen. ‘StartMedia’ represents the time that, for instance, the video-player reports that the video is playing. ‘EndMedia’ is the time that the video-player reports that the video has ended. ‘EndSlide’, finally, represents the point in time where iMotions ended the slide.

Notably, the visualizations in iMotions – for instance in the ‘Replay’ menu – always represent the data from StartMedia until EndMedia, hence, focusing on the time that the ‘slide’ is actually on screen.

Click here to find more information on how to export the ‘sensor data’.

Do you need to care about all this?

Probably not.

Ultimately, the precision of the iMotions software is in the range of tens of milliseconds. That is precise enough for most research purposes supported with iMotions. Consider thereby that the majority of biological responses captured using eye-tracking, GSR, FEA or EEG are themselves much slower than the resolution of the hardware and software. For instance, an ‘arousal peak’, captured by GSR, tends to happen several seconds after an emotionally evoking event, with great case-by-case variability and inter-individual differences. Hence, in the case of GSR-measurements, why would you care about a near-millisecond resolution?

Other users, however, do need to care about a near-millisecond resolution, for instance, researchers interested in Reaction Time (RT) measurements or Event-Related Potentials (ERPs) in EEG.

How can the timestamping be improved further (for a near-millisecond resolution)?

Users who need to care about a near-millisecond resolution have a few more tools available in iMotions to further increase the precision of their measurements.

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Specialized hardware

One solution is to work with specialized hardware, such as photodiodes, trigger-boxes (TTL) and external synchronization units. For the best possible measurement of the onset of a stimulus on screen, for instance, it is possible to work with a photodiode placed in the corner of the screen.

Click here for more information on how to set up the ESU and Stimtracker for ABM B-alert (EEG)

Click here for more information on how to set up the Stimtracker for Brainproducts ActiCHamp (EEG)

API events

It is further possible to live-import ‘events’ through the API-system. If these events-markers are generated on the same computer as the computer used for data-collection (in the iMotions software), the UTC timestamps could be forwarded as part of the event, for a perfect synchronization of the events with the iMotions timeline.

Click here for more information on how to set up communication over the API.

I*n sum: in this article, we have tried to provide an intuition about the many types of latencies that exist in the recording of biometric data, and which make it very difficult to obtain a ‘perfect synchronization’ of different hardware systems. The solution provided by iMotions includes a number of strategies, which minimize these latencies. Ultimately, the standard resolution of the iMotions software is in the range of tens of milliseconds – and importantly, working with the iMotions software does not produce additional delays on top of the latencies that are due to the hardware itself. If necessary, it is possible to further increase this resolution by using the right equipment (e.g., photodiodes, trigger-boxes, etc.). Please reach out to us if, for your research, you need to work with the highest precision; we will help you to find out the best possible solution.*

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